

Solution Set 5

Problem 1: When H^0 includes the Coulomb potential plus the spin orbit coupling we see that the energy eigenstates will be given by

$$|n j m_j; l s\rangle \quad \text{where } l = \text{orbital angular mom} \\ s = \frac{1}{2} \text{ spin angular momentum.}$$

Note that the original degeneracy of the levels, without the spin orbit interaction was $2n^2$. This degeneracy has been lifted by the spin orbit coupling. And we know that levels with a fixed l are split into $j = l + \frac{1}{2}$ and $j = l - \frac{1}{2}$ levels.

We also found (see eq. 15.2.20) of Shankar.

$$|n j = l + \frac{1}{2}, m_j; l s\rangle = \sqrt{\frac{(l + \frac{1}{2} + m_j)}{2l + 1}} |n l m_j - \frac{1}{2}\rangle \otimes | \frac{1}{2} \rangle \\ + \sqrt{\frac{(l + \frac{1}{2} - m_j)}{2l + 1}} |n l m_j + \frac{1}{2}\rangle \otimes | -\frac{1}{2} \rangle.$$

$$|n j = l - \frac{1}{2}, m_j; l s\rangle = - \sqrt{\frac{(l + \frac{1}{2} - m_j)}{2l + 1}} |n l m_j - \frac{1}{2}\rangle \otimes | \frac{1}{2} \rangle \\ + \sqrt{\frac{(l + \frac{1}{2} + m_j)}{2l + 1}} |n l m_j + \frac{1}{2}\rangle \otimes | -\frac{1}{2} \rangle$$

Now adding a weak magnetic field. $\vec{B} = B\hat{k}$ we have

$$H' = \left(\frac{eB}{2mc} \right) (L_z + 2S_z)$$

Since H' operator does not change the " l " quantum #s we just have (we don't have to worry about accidental degeneracies!) and m_j .

$$E' = \langle n j m_j; l s | H' | n j m_j; l s \rangle.$$

$$(L_z + 2S_z) | n j = l \pm \frac{1}{2} m_j; l s \rangle$$

$$= \pm \sqrt{\frac{(l + \frac{1}{2} \pm m_j)}{(2l+1)}} \cdot (m_j \pm \frac{1}{2}) | n l m_j \mp \frac{1}{2} \rangle \otimes | \frac{1}{2} \rangle$$

$$+ \sqrt{\frac{(l + \frac{1}{2} \mp m_j)}{2l+1}} (m_j \mp \frac{1}{2}) | n l m_j \pm \frac{1}{2} \rangle \otimes | -\frac{1}{2} \rangle.$$

$\therefore$

$$E' = \left(\frac{eB}{2mc} \right) \cdot \left\{ (m_j + \frac{1}{2}) \frac{(l + \frac{1}{2} \pm m_j)}{2l+1} + (m_j - \frac{1}{2}) \frac{(l + \frac{1}{2} \mp m_j)}{2l+1} \right\}.$$

$$E' = \left(\frac{eB}{2mc} \right) \left\{ 1 \pm \frac{1}{2l+1} \right\} m_j$$

Problem 2 : There are four degenerate levels $|2, l, m\rangle$. These are $l=0, m=0$, $l=1, m=1, 0, -1$. Let us first construct the 4×4 matrix.

$$\langle 2, l, m | \hat{X} | 2, l', m' \rangle.$$

We see that the non-zero elements are.

$$\langle 200 | \hat{X} | 211 \rangle \text{ and } \langle 200 | \hat{X} | 21-1 \rangle.$$

In fact using Wigner-Eckart theorem alone may indicate

$$\langle 210 | \hat{X} | 211 \rangle \text{ and } \langle 210 | \hat{X} | 21-1 \rangle$$

may be non-zero. But an explicit calculation shows these are zeros.

Given $\langle 200 | \hat{Z} | 210 \rangle = -3a_0$ (from class) we

$$\text{See } \langle 200 | \hat{X} | 211 \rangle = \frac{3}{\sqrt{2}} a_0 = - \langle 200 | \hat{X} | 21-1 \rangle.$$

$$\hat{X} = \begin{matrix} & \begin{matrix} 200 & 210 & 211 & 21-1 \end{matrix} \\ \begin{matrix} 200 \\ 210 \\ 211 \\ 21-1 \end{matrix} & \begin{bmatrix} 0 & 0 & \frac{3a_0}{\sqrt{2}} & -\frac{3a_0}{\sqrt{2}} \\ 0 & 0 & 0 & 0 \\ \frac{3}{\sqrt{2}}a_0 & 0 & 0 & 0 \\ -\frac{3}{\sqrt{2}}a_0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Ignoring $|210\rangle$ we see

$$= \frac{3}{\sqrt{2}} a_0 \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

($|210\rangle$ state does not change!)

The matrix has three vectors:

$$\begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

eigenvalue 0

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

eigenvalue $3a_0$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{2} \\ +\frac{1}{2} \end{pmatrix}$$

eigenvalue $-3a_0$

Zeroth order states are

1st order
energy shift

$$|210\rangle \longrightarrow$$

$$0$$

$$\frac{1}{\sqrt{2}}(|211\rangle + |21-1\rangle) \longrightarrow$$

$$0$$

$$\left(\frac{1}{\sqrt{2}}|200\rangle + \frac{1}{2}|211\rangle - \frac{1}{2}|21-1\rangle \right) \longrightarrow +3a_0\lambda$$

$$\left(\frac{1}{\sqrt{2}}|200\rangle - \frac{1}{2}|211\rangle + \frac{1}{2}|21-1\rangle \right) \longrightarrow -3a_0\lambda$$

Problem 3 : Problem 17.3.4 in Shankar.

(2) $\langle \frac{\lambda}{r^2} \rangle$ is the first order correction to the Hamiltonian if we add $\frac{\lambda}{r^2}$ to the Hamiltonian. As discussed by Shankar we can find the exact answer! by replacing.

$$\frac{\hbar^2 \ell(\ell+1)}{2mr^2} + \frac{\lambda}{r^2} = \frac{\hbar^2 \ell'(\ell'+1)}{2mr^2}.$$

Now $E(\ell') = \frac{-me^4}{2\hbar^2(k+\ell'+1)^2} = E^0 + E' + \dots$

$$\langle \frac{\lambda}{r^2} \rangle = E' = \lambda \left(\frac{\partial E(\ell')}{\partial \lambda} \right)_{\lambda=0} = \lambda \cdot \left(\frac{\partial E}{\partial \ell'} \frac{d\ell'}{d\lambda} \right)_{\lambda=0}$$

$$\langle \frac{\lambda}{r^2} \rangle = \lambda \cdot \left(\frac{me^4}{\hbar^2(k+\ell'+1)^3} \frac{d\ell'}{d\lambda} \right)_{\lambda=0}$$

$$1 = \frac{\hbar^2}{2m} (2\ell'+1) \frac{d\ell'}{d\lambda}.$$

$$\therefore \langle \frac{\lambda}{r^2} \rangle = \lambda \cdot \left(\frac{me^4}{\hbar^2(k+\ell+1)^3} \cdot \frac{2m}{\hbar^2(2\ell+1)} \right) = \frac{\lambda}{\hbar^3 a_0^2 (\ell + \frac{1}{2})}$$

(3) Let us first verify that

$$\frac{P_r^2}{2m} = -\frac{\hbar^2}{2m} \underbrace{\left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right)}_{\left(\frac{2}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2} \right)}$$

where $P_r = -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right)$

$$\begin{aligned} P_r^2 &= -\hbar^2 \left(\frac{\partial^2}{\partial r^2} + \cancel{\frac{1}{r^2}} + \frac{2}{r} \frac{\partial}{\partial r} - \cancel{\frac{1}{r^2}} \right) \\ &= -\hbar^2 \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) \end{aligned}$$

$$\therefore \frac{P_r^2}{2m} = -\frac{\hbar^2}{2m} \left(\frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} \right)$$

Now the Coulomb Hamiltonian can be written as.

$$H = \left(\frac{P_r^2}{2m} + \frac{\vec{L}^2}{2mr^2} - \frac{e^2}{r} \right) \cdot \left\{ \begin{array}{l} \text{see eq.} \\ (12.6.1) \\ \text{and} \\ (12.5.36) \end{array} \right\}$$

$[\vec{L}, P_r] = 0$ since $\vec{L}$ involves angular variables and P_r involves radial variables. Also

$$\left[\frac{1}{r^n}, P_r \right] = -\frac{i\hbar n}{r^{n+1}}$$

$$\therefore [H, P_r] = -i\hbar \left(\frac{\vec{L}^2}{mr^3} - \frac{e^2}{r^2} \right)$$

$$\langle E_n^0 | [H, p_r] | E_n^0 \rangle = E_n^0 \langle p_r \rangle - \langle p_r \rangle E_n^0 = 0.$$

$$\left\langle \frac{\vec{L}^2}{m r^3} \right\rangle = \left\langle \frac{e^2}{r^2} \right\rangle.$$

$$\frac{\hbar^2 l(l+1)}{m} \left\langle \frac{1}{r^3} \right\rangle = \left\langle \frac{e^2}{r^2} \right\rangle$$

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{1}{a_0 l(l+1)} \left\langle \frac{1}{r^2} \right\rangle$$
